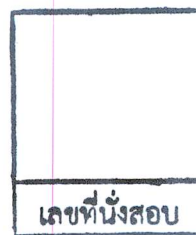


College of Industrial Technology
King Mongkut's University of Technology North Bangkok



Midterm Examination of semester 1

Year: 2017

Subject: 394171 Mathematics I

Section: 15-18

Date 3 October 2017

Time 8:00-10:00

Name _____ ID _____ Field of study _____

Instructions

1. The examination has 8 pages (including this page) and a total score of 90 points.
2. Write all your solutions and answers on this examination sheet.
3. This is a closed book examination.
4. You are not allowed to leave the examination room during the first 1 hour after the beginning of the examination.
5. You are not allowed to open the exam papers or start to answer before the proctor's permission.
6. You are not allowed to use the restroom during the exam except in case of an emergency.
7. No documents are allowed to be taken out of the examination room.
8. Calculators are not allowed in the examination.
9. Electronic communication devices are **NOT** allowed in the examination room.

Cheating In the exam is considered an extremely serious offence which will result in expulsion from the University

Question 1. Given $A = \{ 1, 2, -3, 4, \{1, 3\}, \{1\}, \{1, \{1, 2\}\} \}$, determine the following expressions True or False. (10 marks)

- | | |
|------------------------------------|------------------------------|
| 1. $12 \in A$ | 2. $\{1, 2\} \in A$ |
| 3. $\{1\} \notin A$ | 4. $\{1, 2, 3\} \notin A$ |
| 5. $\{1, 3\} \subset A$ | 6. $\{1, -3\} \subset A$ |
| 7. $\{1, \{1, 3\}\} \not\subset A$ | 8. $\{3, -3\} \not\subset A$ |
| 9. $\{1, \{1, 3\}\} \in P(A)$ | 10. $\{\{3\}\} \subset P(A)$ |

Question 2. Given $A \cup B = \{1, 2, 3, 4, 5, k\}$ and $A \cup C = \{1, 2, 3, 4, 6, 7, 8, k\}$
 $A \cap C = \emptyset$, $B \cap C = \{k\}$, $A \cap (B \cup C) = \{1, 2\}$. Find the sets of A and B.
 (10 marks)

Question 3. In a survey asking student for which subject they want to study, there are 20 students who want to study in Mathematics or Physics classes . If the students enroll Physics class, then they will not enroll Mathematics class. There are 17 students who do not enroll Mathematics class. There are 15 students who do not enroll Physics class. How many students who do not enroll both classes? (10 marks)

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Question 4. 4.1 Find the set of solutions of $\frac{x}{2} \geq \frac{x}{x^2-1}$. (5 marks)

4.2 Let B be a set of solutions of $3x^2 - 5x^{-1} - 2 \leq 0$. Evaluate the interval of $[-2, 4) \cap B$. (5 marks)

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Question 5. Let a and b be any nonnegative real numbers and the operator is defined by $a * b = a + b + 2\sqrt{ab}$

5.1 Evaluate the value of $0 * (4 * 1)$. (3 marks)

5.2 Evaluate the identity of the operator $a * b = a + b + 2\sqrt{ab}$ (7 marks)

Question 6. Let x and y be any real numbers such that $-6 \leq x \leq -1$ and $2 \leq y \leq 8$,
(10 marks)

6.1 Evaluate the intervals of $x + y$,

6.2 Evaluate the intervals of $x - y$,

6.3 Determine the interval of xy .

6.4 Determine the interval of $\frac{x+y}{y}$.

Question 7. 7.1 Find the remainder of the expression $\frac{x^5 + 6x^4 - 4x^3 - x + 2}{2x - 1}$.
(4 marks)

7.2 Find a set of the solution of $\frac{(x^2 + 2x - 3)\sqrt{x - 4}}{|2x - 1|} \leq 0$. (6 marks)

Question 8. Given the relation $r = \left\{ (x, y) \in \mathbb{R} \times \mathbb{R} \mid y = \frac{1-2\sqrt{x}}{\sqrt{x}-2} \right\}$,

Determine the set of D_r and $D_r \cap R_r$. (10 marks)

Question 9. Given the two relations $r = \left\{ (x, y) \in \mathbb{R} \times \mathbb{R} \mid \sqrt{x} < y \right\}$
and $s = \left\{ (x, y) \in \mathbb{R} \times \mathbb{R} \mid x < y^2 \right\}$. Sketch the graphs of both relations and show that $r \subset s$. (10 marks)